Object Detection Extension

https://github.com/aiyshwary/Object-Detection-Extension-in-Rapid-Miner

OVERVIEW

A RapidMiner Studio extension that brings transfer learning-based object detection into a no-code enviro
inference with Faster R-CNN, FCOS, SSD, SSDLite, and RetinaNet models.

IMPACT

Developed two RapidMiner operators for object detection — FineTuneObjectDetectionModel and ObjectD
users to train and deploy detection models within a visual workflow without writing code.

KEY DETAILS

1. FineTuneObjectDetectionModel accepts image and annotation directories (YOLO/Pascal VOC) and ou
CSV.
2. ObjectDetectionModelInference runs predictions on new images and returns bounding boxes and clas
3. Supports five architectures: Faster R-CNN, FCOS, SSD, SSDLite, and RetinaNet for varying speed-ac
4. Configurable for CPU-only and GPU (CUDA cu118) environments; installed as a JAR extension in Ra
5. Enables domain-specific object detection with minimal ML expertise using a drag-and-drop process de
6. Inference operator can optionally save annotated images with overlaid prediction bounding boxes via N
output.
7. Uses chardet for automatic annotation-file encoding detection, ensuring robust parsing of YOLO/Pasca
systems and locales.